

Curriculum Vitae

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PhD Advisors: Prof. Jinsheng Cai (The Yangtze River Scholar)



Work experience

- **2024/12-Now**, City University of Hongkong (CityU), Department of Architecture and Civil Engineering, Research Associate.

Education Profile

- **2020/03-2024/12**, Northwestern Polytechnical University (NPU), School of Aeronautics, Fluid Mechanics, PhD.
- **2017/09-2020/03**, Northwestern Polytechnical University (NPU), School of Electronics and Information, Electromagnetic Field and Microwave Technology, Master.
- **2018/07-2018/08**, Novosibirsk State University, Russia, Visiting student.
- **2013/09-2017/06**, Northwestern Polytechnical University (NPU), School of Electronics and Information, Electromagnetic Field and Microwave Technology, Bachelor.

Research Interests

- **Aerodynamic:** Turbulence drag reduction, Flow control, Computational fluid dynamics, Machine learning in fluid mechanics, etc.
- **Built Environment:** Simulation of fluid mechanics in built environment; Efficient and physics informed surrogate model of fluids.

Research Papers

1. **Yue Zhang**, Xing Zheng*. Influence of geometric parameters of façade protruding ribs on turbulent flow statistics in street canyons: A large-eddy simulation study. *Building and Environment*. 2026, 287, 113845.
2. **Yue Zhang**, Xing Zheng*. POD-Kriging surrogate modeling for rapid prediction of the impact of façade protruding rib geometries on urban canyon wind flow. *Building and Environment*. 2026, 292, 114198.
3. **Yue Zhang**, Shaohui Zhang, Fei Jiang, Zhaoqian Wu, Weihua Li, Predicting surface pressure fields and lift coefficients of subsonic airfoils by machine-learning-enhanced compressive sensing framework. *Physics of Fluids*, 2025, 37, 095103.
4. **Yue Zhang**, Cai jinsheng, Wenfeng Li*. A framework of the effective simulation of high-Reynolds number turbulent flows over ribbed surfaces using a data-driven boundary surrogate model. *AIAA Journal*. 2026 64:2, 808-822.
5. **Yue Zhang**, Cai jinsheng, Wenfeng Li*. Direct numerical simulations of the drag degradation mechanism in channel flow over trapezoidal riblets. *Aerospace Science and Technology*. 2024, 144, 108821
6. **Yue Zhang**, Fei Jiang, Zhaoqian Wu, Zhi Zhong, Shaohui Zhang, Weihua Li, Direct numerical

simulations of streamwise-aligned riblets: Investigations of the blockage ratio. *Physics of Fluids*, 2026, 38(7), 075184.

7. **Yue Zhang**, Jinsheng Cai and Wenfeng Li*, Turbulent Channel Flow With Riblets In Drag-Reducing And Drag-Increasing Regimes, AIAA Aviation 2022, AIAA 2022-3851. (High-level international conference, EI)
8. **Yue Zhang***, Chenjiang Guo, An UWB Dual-Polarized Low Profile Tight Coupled Dipole Array, 2019 International Applied Computational Electromagnetics Society Symposium - China (ACES). (EI)
9. 许德辰, **张悦**, 钱战森, 等. 飞艇微肋条减阻模化方法研究[J]. 气动研究与试验, 2026, 4(1): 33–41.
10. 许德辰, **张悦**, 刘欣乐, 李文丰. 基于粒子追踪测速的壁面摩擦应力测量[J]. 实验流体力学, 2022, 36(2): 131–138.

Conference

- Research on Riblet Modeling Method and Application Based on Slip Boundary, *12th National Conference on Fluid Mechanics*, Xi'an, Shaanxi, China. (**Best Poster Presentation**)
- Experimental Study of the Effect of Wall Microstructure on the Streamwise Development of Turbulent Boundary Layer, *11th National Conference on Fluid Mechanics*, Shenzhen, Guangdong, China.
- Turbulent Channel Flow with Riblets in Drag-Reducing and Drag-Increasing Regimes, *AIAA AVIATION 2022 Forum*, Chicago, USA. (Oral Presentation)

Project Experiences

- “**Anti-icing and drag-reducing composite surface for civil aircraft components**”, Civil aircraft special research project of Ministry of Industry and Information Technology, research fund 2 million yuan. Participants.
- “**Boundary layer drag reduction control mechanism and design evaluation**”, cooperative project with 626 research institution, research fund 0.6 million yuan. Participants.
- “**Research on active and passive coupling control method of boundary layer drag reduction and flow mechanism**”, National Natural Science Foundation of China, research fund 0.3 million yuan. Participants.

Language and Software skills

- CET 6: 544 / TOEFL: 91
- Software: ANSYS Fluent / OpenFOAM / Pointwise / ICEM / Paraview / Tecplot / Matlab / Python / ANSYS HFSS etc.